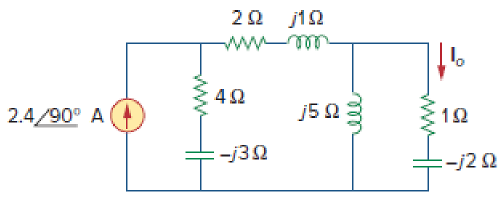
Problem

Find **I**o in the circuit of Fig. 10.19 using the concept of source transformation.

Figure 10.19:



Step-by-step solution

Step 1 of 5

Refer the figure 10.19 in the text book.

First transform the current source into a voltage source. Determine the value of voltage source.

$$\begin{aligned} V_s &= (2.4\angle 90°)(4 - j3) \\ &= (2.4\angle 90°)(5\angle -36.87^0) \\ &= 12\angle 53.13° \text{ V} \end{aligned}$$

Re-draw the circuit diagram:

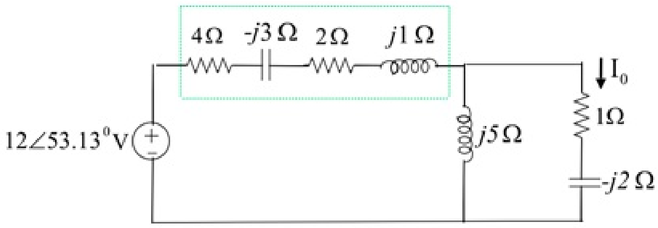


Figure 1

Step 2 of 5

Determine the series combination of $4-j3$ and $2+j$ impedances.

$$Z_1 = 4 - j3 + 2 + j$$
$$= (6 - j2)\Omega$$

Convert the voltage source to a current source. Determine the value of current source.

$$I_s = \frac{V_s}{Z_1}$$
$$= \frac{12\angle 53.13°}{6 - j2}$$
$$= 1.9\angle 71.56°\ \text{A}$$

Re-draw the circuit diagram.

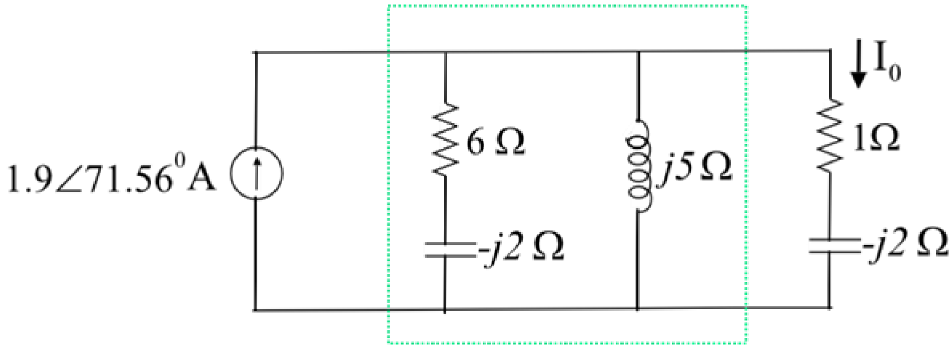


Figure 2

Step 3 of 5

Determine the parallel combination of $6-j2$ and $j5$ impedances.

$$Z_2 = \frac{(6 - j2)\,j5}{6 - j2 + j5}$$
$$= \frac{10 + j30}{6 + j3}$$
$$= 4.71\angle 45°$$
$$= 3.33 + j3.33\ \Omega$$

Step 4 of 5

Re-draw the simplified circuit as shown in figure 3.

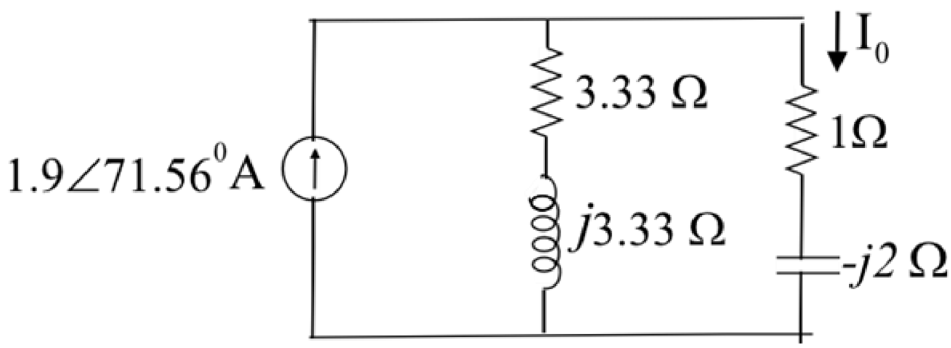


Figure 3

Step 5 of 5

From Figure 3, apply current division rule and determine the value of I_0.

$$
\begin{aligned}
I_0 &= 1.9\angle 71.56^\circ \times \frac{3.33 + j3.33}{3.33 + j3.33 + 1 - j2} \\
&= 1.9\angle 71.56^\circ \times \frac{3.33 + j3.33}{4.33 + j1.33} \\
&= 1.9\angle 71.56^\circ \times \frac{4.71\angle 45^\circ}{4.53\angle 17.07^\circ} \\
&= 1.97\angle 99.49^\circ \text{ A}
\end{aligned}
$$

Therefore, the value of current I_0 is $\boxed{1.97\angle 99.49^\circ \text{ A}}$.